DVN36104 SPR 2026

Data Visualisation and Narratives
Spring 2026
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36104 · DATA VISUALISATION AND NARRATIVES

Color and Perception

How many colours are there?

an English speaker’s basic colour words	11
the classic Crayola box	64
the web’s named colours	148
codes on a modern screen	16,777,216
differences a human can distinguish	a few million
the spectrum	a continuum

The web’s named colours

The 148 CSS colour keywords, inherited from the graphical display protocol X11 (recognised by matplotlib).

aliceblue antiquewhite aqua aquamarine azure beige bisque black blanchedalmond blue blueviolet
brown burlywood cadetblue chartreuse chocolate coral cornflowerblue cornsilk crimson cyan
darkblue darkcyan darkgoldenrod darkgray darkgreen darkgrey darkkhaki darkmagenta
darkolivegreen darkorange darkorchid darkred darksalmon darkseagreen darkslateblue
darkslategray darkslategrey darkturquoise darkviolet deeppink deepskyblue dimgray dimgrey
dodgerblue firebrick floralwhite forestgreen fuchsia gainsboro ghostwhite gold goldenrod gray
green greenyellow grey honeydew hotpink indianred indigo ivory khaki lavender lavenderblush
lawngreen lemonchiffon lightblue lightcoral lightcyan lightgoldenrodyellow lightgray
lightgreen lightgrey lightpink lightsalmon lightseagreen lightskyblue lightslategray
lightslategrey lightsteelblue lightyellow lime limegreen linen magenta maroon mediumaquamarine
mediumblue mediumorchid mediumpurple mediumseagreen mediumslateblue mediumspringgreen
mediumturquoise mediumvioletred midnightblue mintcream mistyrose moccasin navajowhite navy
oldlace olive olivedrab orange orangered orchid palegoldenrod palegreen paleturquoise
palevioletred papayawhip peachpuff peru pink plum powderblue purple rebeccapurple red
rosybrown royalblue saddlebrown salmon sandybrown seagreen seashell sienna silver skyblue
slateblue slategray slategrey snow springgreen steelblue tan teal thistle tomato turquoise
violet wheat whitesmoke yellow yellowgreen

English has eleven basic colour terms

black	white	red	green
yellow	blue	brown	purple
pink	orange	grey	

A basic term is a single word, not derived from an object (unlike “gold” or “turquoise”), and applies to a broad class of things.

Berlin and Kay (1969), Basic Color Terms.

Priority of Colour Words

Stage	Terms present
I	black white
II	+ red
III–IV	+ green and yellow, in either order
V	+ blue
VI	+ brown
VII	+ purple pink orange grey

Languages differ in how many basic terms they have; acquisition follows this order with few exceptions. Red is the first hue named everywhere.

The classic 64-colour Crayola box

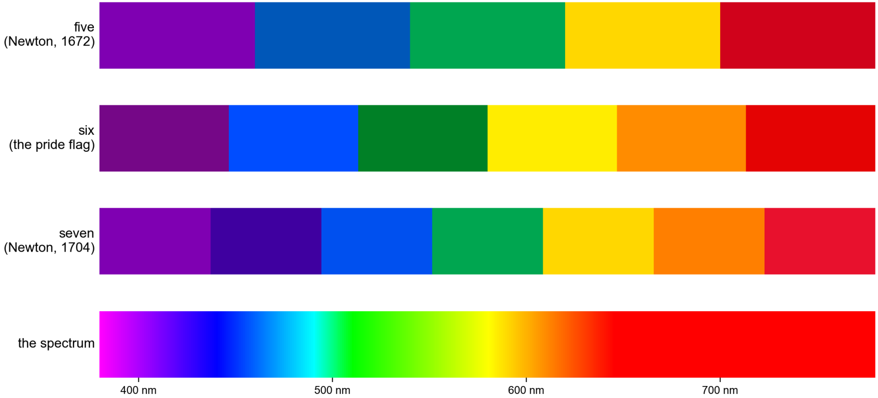
Apricot
Asparagus
Bittersweet
Black
Blue
Blue Green
Blue Violet
Bluetiful
Brick Red
Brown
Burnt Orange
Burnt Sienna
Cadet Blue
Carnation Pink
Cerulean
Chestnut
Cornflower
Forest Green
Gold
Goldenrod
Granny Smith Apple
Gray

Green
Green Yellow
Indigo
Lavender
Macaroni and Cheese
Magenta
Mahogany
Mauvelous
Melon
Olive Green
Orange
Orchid
Pacific Blue
Peach
Periwinkle
Plum
Purple Mountains' Majesty
Raw Sienna
Red
Red Orange
Red Violet

Robin's Egg Blue
Salmon
Scarlet
Sea Green
Sepia
Silver
Sky Blue
Spring Green
Tan
Tickle Me Pink
Timberwolf
Tumbleweed
Turquoise Blue
Violet (Purple)
Violet Red
White
Wild Strawberry
Wisteria
Yellow
Yellow Green
Yellow Orange

Product image: Crayola.

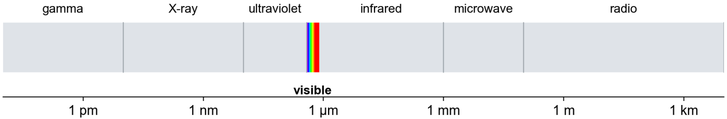
Colours of the rainbow



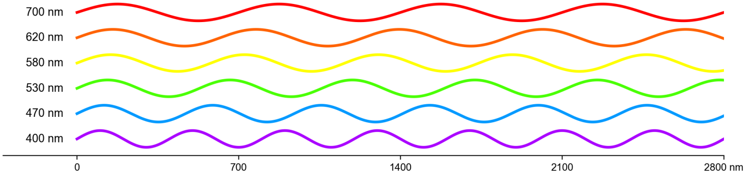
The band is continuous; the number of named colours is a choice. Newton first cut it into five, then seven, to match the notes of the musical scale.

Visible light is a narrow band

The electromagnetic spectrum, by wavelength



Six visible wavelengths, drawn to scale



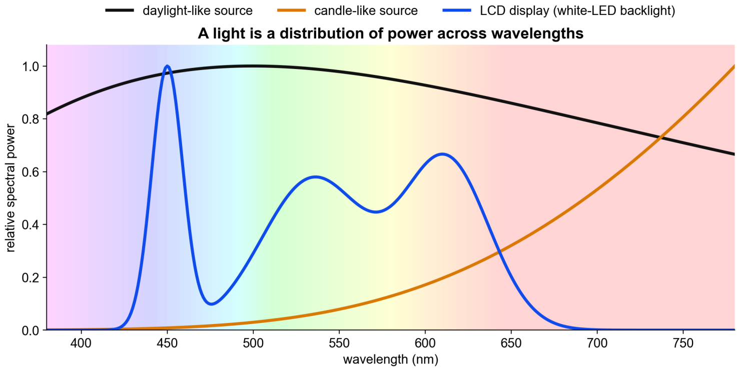
Light is electromagnetic radiation. The visible band runs from about 380 to 780 nanometres, between ultraviolet and infrared.

Rods and cones

	Rods	Cones
count	about 120 million	about 6 million
work in	dim light	daylight
types	one	three
colour signal	none	three compared responses
concentration	periphery	packed in the fovea

At chart-reading light levels the rods are saturated.

Spectral recipes of common lights



Course calculation. Black-body curves are illustrative; the display curve is modelled on a white-LED LCD.

A light is a function

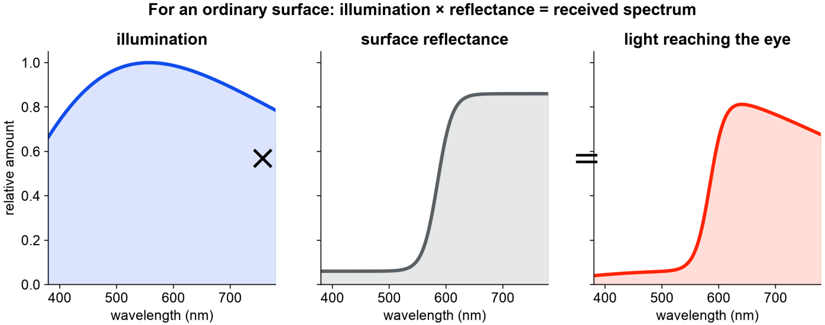
$$P(\lambda) = \text{power at wavelength } \lambda$$

A spectral power distribution contains one value for every wavelength sampled.

- a narrow peak: approximately monochromatic;
- a broad curve: many wavelengths;
- three peaks: a typical display mixture;
- one colour word: none of the above is specified.

The spectrum is the physical input; appearance comes several steps later.

A surface is also a function



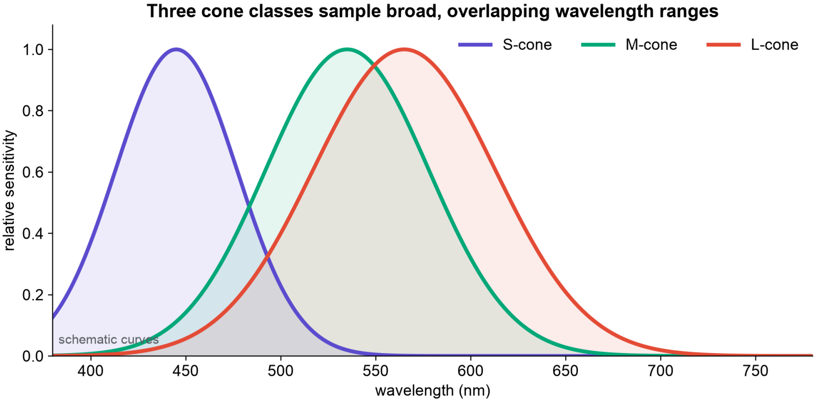
A surface reflects some fraction of the light at each wavelength. That fraction, between 0 and 1, is its **reflectance** $R(\lambda)$. A pigment is a material that sets this curve. The light reaching the eye is the product $P(\lambda)R(\lambda)$, so the same surface sends a different spectrum under a different light. Mixing pigments multiplies their curves, and multiplying fractions always gives less: pigment mixtures darken.

Which layer does each statement describe?

- “This source has a peak at 470 nm.”
- “This surface reflects more short-wavelength light under this lamp.”
- “This patch appears blue in this surround.”
- “This pixel is encoded as `#0068B7` in sRGB.”

Each statement describes one layer. Knowing one layer does not fix the other three.

Three cone classes sample overlapping ranges



Schematic curves for teaching. S, M, and L mean short-, middle-, and long-wavelength sensitive.

One cone cannot identify a wavelength

$$q_i = \int P(\lambda) s_i(\lambda)\, d\lambda$$

Multiply the incoming spectrum by one sensitivity curve, then add over wavelength.

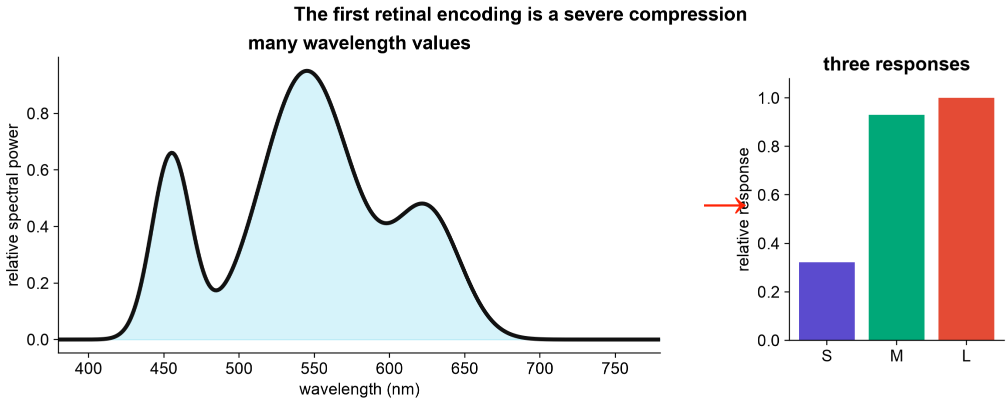
This is the principle of univariance.

The same response can be caused by:

- a weak light near peak sensitivity;
- a stronger light farther from the peak;
- many mixtures across the spectrum.

Once absorbed, the photon carries no wavelength label in that cone's output.

A spectrum is compressed to three responses



The bucket analogy

Useful part

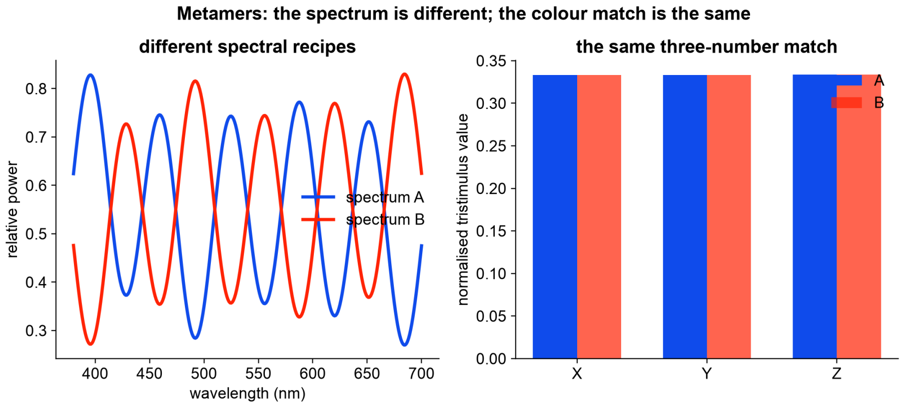
- three buckets have different filters;
- each produces only one total;
- the original spectrum cannot be reconstructed from three totals.

Where it breaks

- cone sensitivities overlap broadly;
- responses adapt and saturate;
- later neurons compare cones;
- context and spatial structure alter appearance.

Three cone classes explain the dimensionality of colour *matching*. They do not, by themselves, explain all colour appearance.

Different spectra can produce the same match



Course construction using the CIE 1931 colour-matching functions.

The same code can look different

Identical centre RGB values do not guarantee identical appearance. The visual system compares each patch with its surround.

After the cones, signals are compared

luminance-like	$L + M$
red–green-like	$L - M$
blue–yellow-like	$S - (L + M)$

These are schematic combinations, far from a complete neural model. They show that the brain does not receive an RGB image from the retina.

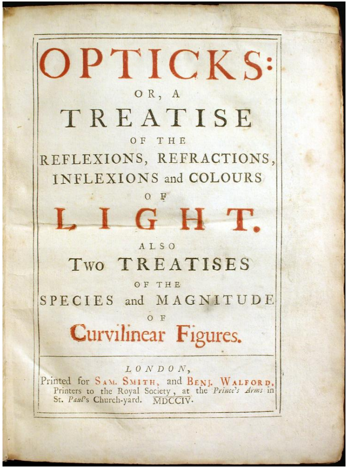
Adaptation, local contrast, spatial scale, and viewing sequence all enter before conscious appearance.

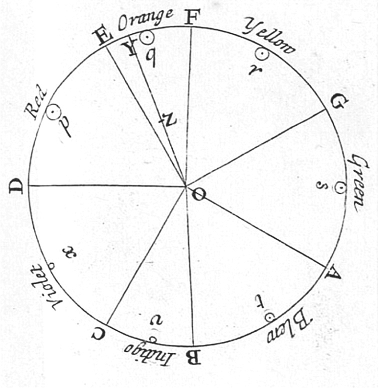
Eight questions about colour, 1704 to 1931

Work	Object	Question
Newton, 1704	rays and prisms	How does white light separate and recombine?
Harris, 1766	pigment mixtures	How can pigment mixtures be ordered on one wheel?
Dalton, 1794	his own vision	How can colour blindness be described and explained?
Goethe, 1810	appearance	How do boundaries, shadows, afterimages, and context affect experience?
Maxwell, 1850s–60s	matches	How can three primaries reproduce a match?
Hering, 1878	structure of appearance	Which hues look unmixed, and which blends never appear?
Munsell, 1905	surface samples	How can hue, value, and chroma be ordered for practice?
CIE, 1931	standard observer	How can colour matches be reported reproducibly?

The 1931 system settles the matching question only. The earlier questions about appearance and practice stay open.

Newton, Opticks (1704)





Newton's colour circle, Opticks, Book I, Part II. Public domain.

Newton and Goethe

Newton

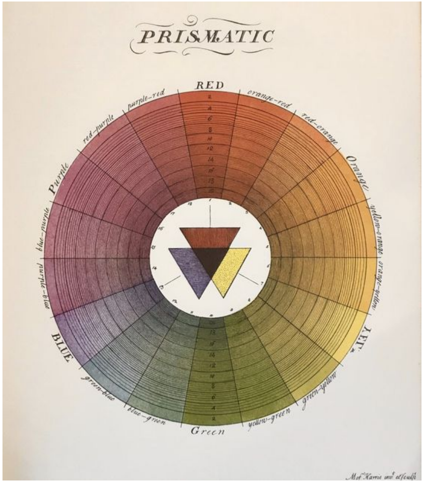
- controlled the geometry;
- separated light by refraction;
- showed white light can be compound;
- described the stimulus.

Goethe

- studied coloured shadows;
- recorded afterimages and boundaries;
- foregrounded the observer;
- described appearance in context.

Goethe’s physical theory was wrong; Newton’s optics stands. His records of shadows and afterimages still show that appearance depends on the observer and the surround.

Harris, 1766

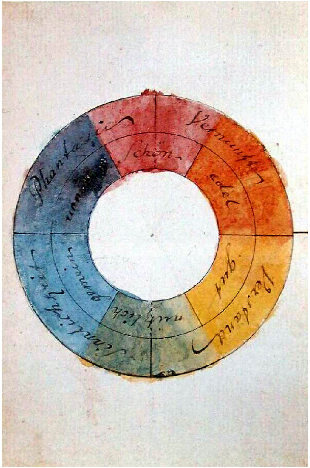


Moses Harris, an engraver and entomologist, published the first colour-printed colour wheel in *The Natural System of Colours*. Red, yellow and blue sit at the centre as the “grand or principal” colours, with mixtures grading around the circle.

The school story that red, yellow and blue are the three primary colours descends from this wheel and the painters’ practice behind it.

M. Harris, The Natural System of Colours (c. 1766). Public domain.

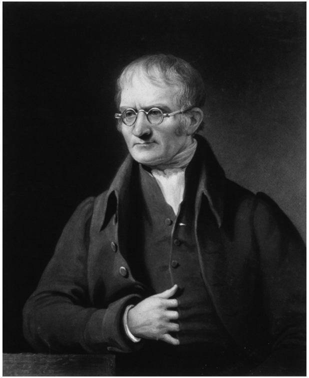
Goethe, Farbenkreis (1809)



Goethe's watercolour wheel assigns qualities to colour sectors. As physics it failed; as a record of how colours are experienced and opposed in pairs, it anticipated opponent descriptions of appearance.

J. W. von Goethe, 1809 watercolour. Public domain, via Wikimedia Commons.

Dalton, 1794



Mezzotint by C. Turner, 1834. Public domain.

Dalton gave the first scientific self-report of colour blindness: to him, “that part of the image which others call red appears to me little more than a shade”. He blamed a blue tint in his eye’s fluid.

He was wrong about the cause. DNA from his preserved eye (1995) showed he lacked the M-cone pigment: one of the three integrators from Part II was missing, so his matches needed only two primaries.

Colour vision deficiencies

Type	Affected pigment	Rate
deutan	M shifted or missing	about 6% of men
protan	L shifted or missing	about 2% of men
tritan	S affected	about 1 in 10,000, both sexes
monochromacy	two or three pigments	very rare

The red–green forms are X-linked, so they concentrate in men: about 8% of men and 0.5% of women. In a class of two hundred students, expect eight or nine.

The colour bars, simulated

Under deuteranopia the green, red and magenta bars converge. A chart has to stay readable in all four panels.

Hering, 1878

Photograph, public domain, via Wikimedia Commons.

Hering organised appearance around three opponent pairs: black–white, red–green, yellow–blue. The four chromatic poles are the **unique hues**: a red that looks neither yellowish nor bluish, and likewise green, yellow and blue.

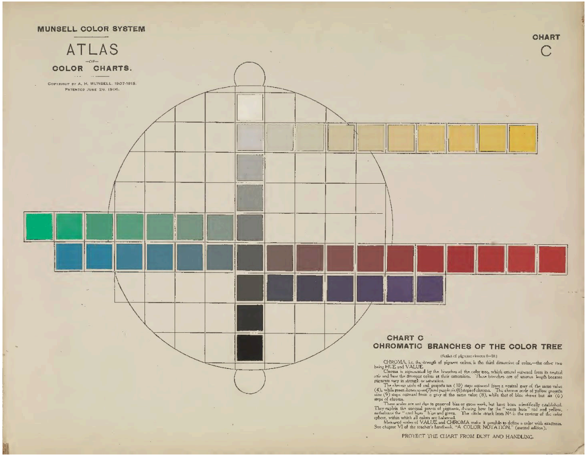
The test is in experience itself: reddish-yellow and greenish-blue exist, and reddish-green and yellowish-blue never appear. Hurvich and Jameson measured the pairs by hue cancellation in 1957.

The opponent cone combinations shown earlier are related but do not line up exactly with the unique hues; their neural basis is still argued. Hering's six poles are also Berlin and Kay's first six basic colour terms.

The unique-hues argument, in short

1. Some hues look like mixtures: orange looks like red plus yellow.
2. Four do not: red, green, yellow and blue look unmixed.
3. Some blends never appear: nothing ever looks reddish-green or yellowish-blue.
4. So appearance runs on two hue axes, red–green and yellow–blue, with an unmixed pole at each end.

Munsell, 1905



A. H. Munsell, Atlas of the Munsell Color System (1907–1915), Chart C. Public

Munsell ordered painted samples along three dimensions: hue around a circle, value (lightness) up the trunk, chroma outward along the branches. Steps were chosen to look equal to observers.

The branches have unequal lengths because pigments reach different maximum chroma at each hue and value: on this chart red extends ten steps, blue only six.

Soil science still uses it; later "perceptually uniform" colour spaces descend from it.

Why a pure cyan cannot be matched

	S	M	L
target: pure cyan, 490 nm	6	10	**4**
green lamp, 546 nm, per unit	0	10	**8**
blue lamp, 436 nm, per unit	10	1	0.5
red lamp, 700 nm, per unit	0	0	2

Illustrative numbers. Reaching M = 10 takes one unit of green, and that unit already delivers L = 8. The cyan needs L = 4, with the red knob already at zero. The green lamp is also a red-cone lamp.

One move remains: add red to the *test* side until its L rises to meet the mix.

Colour matching came before direct cone measurement

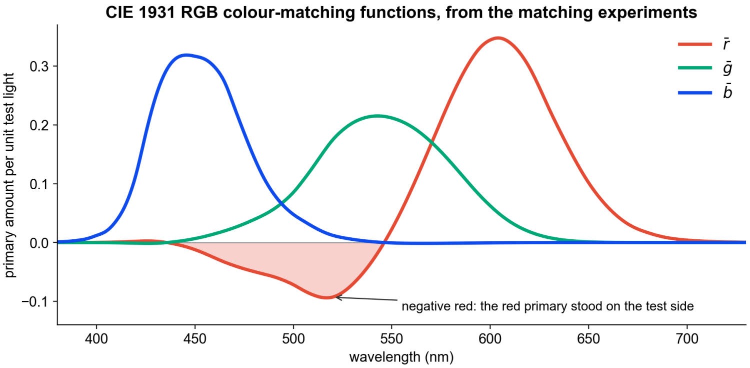
An observer adjusted three primary lights until one half of a field matched a test light in the other half.

Some wavelengths could not be matched using positive amounts of the chosen real primaries. One primary was moved to the test side:

$$\text{test} + rR = gG + bB \quad \Longleftrightarrow \quad \text{test} = (-r)R + gG + bB.$$

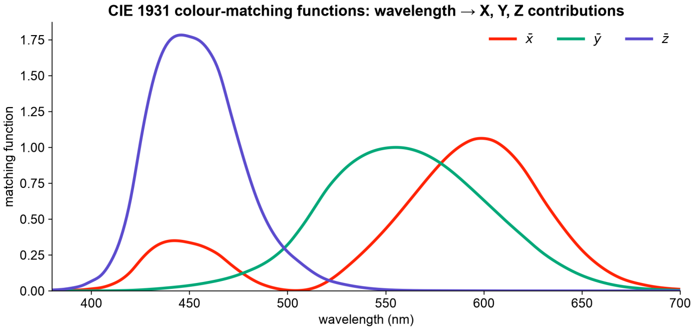
The negative value records the experimental operation. It is not “negative light.”

The measured curves contain negative red



Each curve records how much of one primary matched a unit of single-wavelength test light. In the negative region the red primary stood on the test side of the field.

CIE XYZ is a mathematical coordinate system



XYZ is a linear transform of the curves on the previous slide, chosen so that no value is negative. X, Y, and Z are not physical lamps and not the three cone signals; Y was chosen to carry standard photopic luminance. The second bump of $\bar{x}$ in the violet is real: violet also excites the red-sensitive mechanism, which is why violet looks reddish.

From a spectrum to three CIE values

$$X = \int P(\lambda)\bar{x}(\lambda)\, d\lambda$$

$$Y = \int P(\lambda)\bar{y}(\lambda)\, d\lambda$$

$$Z = \int P(\lambda)\bar{z}(\lambda)\, d\lambda$$

Change the spectrum and recalculate. Any spectra with the same XYZ values are a standard-observer colour match under the specified conditions.

Chromaticity removes one scale factor

$$x = \frac{X}{X+Y+Z}, \qquad y = \frac{Y}{X+Y+Z}, \qquad z = 1 - x - y$$

- Multiplying X, Y, and Z by the same amount leaves x and y unchanged.
- Two fractions are enough because the three proportions sum to one.
- xy omits overall tristimulus scale. It is not a complete colour appearance.

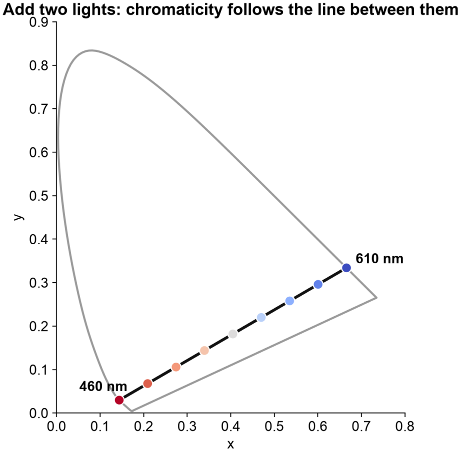
Build the spectral locus one wavelength at a time

For each row in the official CIE table:

1. take the wavelength's $\bar{x}$, $\bar{y}$, and $\bar{z}$ values;
2. divide by their sum;
3. plot (x, y);
4. repeat for the next wavelength.

The curved path is the **spectral locus**: the route taken by monochromatic lights through the xy diagram.

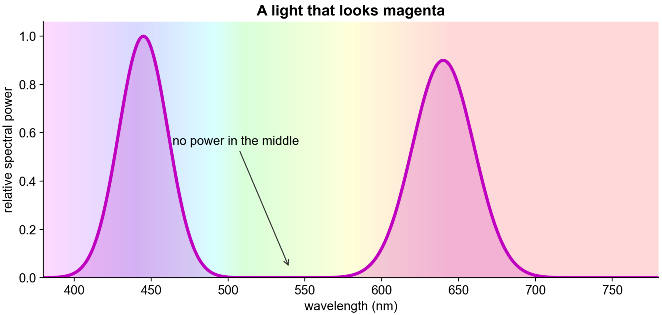
Adding two lights traces a straight line



Why real-light mixtures lie inside

- spectra add;
- XYZ values therefore add;
- after normalisation, a two-light mixture lies between the two chromaticities;
- a spectrum with many wavelengths is a non-negative mixture of many points.

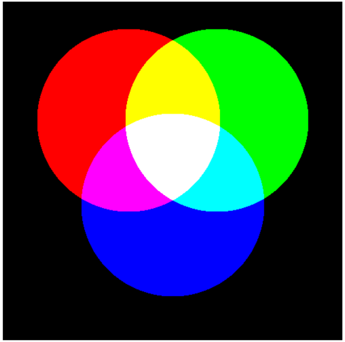
Nonspectral colours



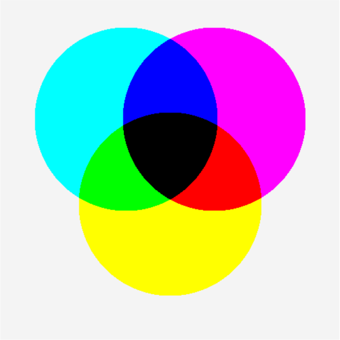
The straight edge joining the two ends of the horseshoe is the **line of purples**; no single wavelength sits on it. A light that looks magenta holds power near both ends of the spectrum, because its cone signature (S and L strong, M weak) is one no single wavelength can produce. Blue names a band of the rainbow; magenta names only a mixture.

Additive and subtractive mixing

Add light: channel powers add



Stack ideal filters: transmissions multiply



Additive mixing combines emitted spectral power. Ideal subtractive mixing multiplies transmissions; real pigments also scatter and interact.

RGB and CMYK

	RGB	CMYK
where	screens	ink on paper
mixing	additive: channels add light	subtractive: inks absorb light
primaries	red, green, blue	cyan, magenta, yellow, black
all zero	black (screen off)	white (bare paper)
all full	white	muddy near-black
you meet it as	`#RRGGBB` in CSS and charts	print exports and PDF proofs

The K in CMYK is a separate black ink, the printers' "key" plate. Full cyan, magenta and yellow together make a muddy brown and soak the paper, so text and shadows use black ink instead. Screens and printers also cover different gamuts: some screen colours have no printable match, worth checking before a chart goes to print.

Red, yellow, blue

- The school primaries are the painters' tradition, running from Le Blon's three-plate printing (1725) through Harris's wheel (1766).
- Printing replaced them with cyan, magenta and yellow. Each ink absorbs one band: cyan absorbs long wavelengths, magenta middle, yellow short. Historical red and blue pigments behaved like impure magenta and impure cyan, which is why the triad worked at all.
- The tell is purple: true red plus true blue passes almost no light and mixes mud. Painters mixed purples from crimson, a magenta-leaning red. Magenta plus cyan mixes purple cleanly.
- Red, yellow and blue also look unmixed (orange looks like a mixture; red does not). Green looks unmixed too, but paint can mix it, so it lost its place in the triad.
- Light mixing uses red, green and blue, matched to where the cones separate best.
- No three colours are "the" primaries: the number three comes from the cones, the choice of three from the medium.

Metamerism is what makes displays possible

A screen does not reconstruct the original spectral recipe. It emits a new spectrum from three primaries that is intended to produce the required match.

What this makes possible

- colour photography;
- television and computer displays;
- compact three-channel specifications.

Where a match can fail

- a different observer;
- a different illuminant;
- fluorescence;
- changed adaptation or field size.

A white LED is a blue LED plus a phosphor

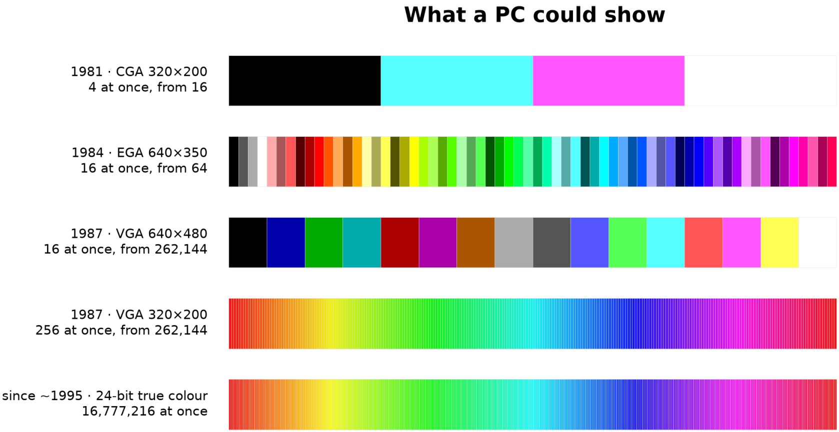
1. The backlight is a **blue LED** emitting a narrow spike near 450 nm.
2. A **phosphor** (typically YAG:Ce) absorbs part of the blue and re-emits it at lower energies: a broad hump across green, yellow and red.
3. Spike plus hump reads as white. The LCD’s red and green filters carve their two humps out of the phosphor light; the blue filter passes the spike.

The blue spike stays tallest in the power spectrum because it is the only light generated directly; everything else is converted from it, with losses.

The blue LED came thirty years late

- An LED emits photons with the energy of its semiconductor's band gap. Red photons need about 1.9 eV; blue photons need about 2.8 eV.
- Red LEDs existed by 1962 (gallium arsenide phosphide). Blue required a wide-gap crystal, in practice gallium nitride.
- Gallium nitride had two standing problems: no matching substrate to grow clean crystals on, and no working p-type doping.
- Akasaki, Amano and Nakamura solved both around 1986–1992 (a buffer layer for growth; activated magnesium doping). Bright blue LEDs followed in 1993, white LEDs in 1996.
- Nobel Prize in Physics, 2014. Every white LED screen and lamp descends from that blue chip.

What a PC could show



The default 16-colour VGA palette

Black
`(0, 0, 0)`
`#000000`
	Blue
	`(0, 0, 170)`
	`#0000AA`
		Green
		`(0, 170, 0)`
		`#00AA00`
			Cyan
			`(0, 170, 170)`
			`#00AAAA`

Red
`(170, 0, 0)`
`#AA0000`
	Magenta
	`(170, 0, 170)`
	`#AA00AA`
		Brown
		`(170, 85, 0)`
		`#AA5500`
			Light grey
			`(170, 170, 170)`
			`#AAAAAA`

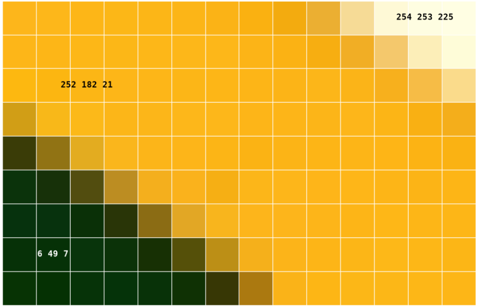
Dark grey
`(85, 85, 85)`
`#555555`
	Light blue
	`(85, 85, 255)`
	`#5555FF`
		Light green
		`(85, 255, 85)`
		`#55FF55`
			Light cyan
			`(85, 255, 255)`
			`#55FFFF`

Light red
`(255, 85, 85)`
`#FF5555`
	Light magenta
	`(255, 85, 255)`
	`#FF55FF`
		Yellow
		`(255, 255, 85)`
		`#FFFF55`
			White
			`(255, 255, 255)`
			`#FFFFFF`

A picture is a grid of numbers

the picture

the marked region: 14 × 9 pixels



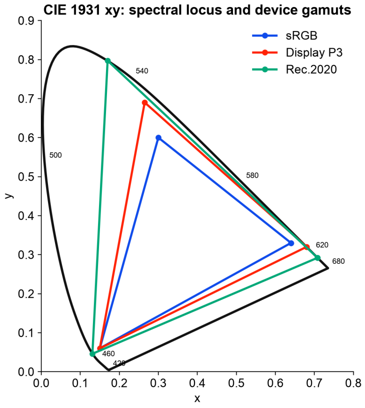
Zoom far enough into any picture and it becomes a grid of squares: **pixels**. A **bitmap** is that grid stored as numbers, three per pixel: how much red, green and blue light to emit, each from 0 to 255. The near-white pixel is high in all three channels; the yellow one is red plus green with almost no blue.

Bits per pixel

Bits per pixel	Colours	Era
1	2	terminals, early Macintosh
2	4	CGA graphics, 1981
4	16	EGA and VGA high-resolution modes
8, indexed	256	VGA 320×200 mode, 1987
15 or 16	32,768 or 65,536	high colour, early 1990s
24	16,777,216	true colour, since the mid-1990s

24 bits is the `#RRGGBB` code: two hexadecimal digits per channel. Chart libraries, CSS and image formats all speak it.

Three physical primaries make a triangle



CIE 1931 2° data; primary coordinates from sRGB/Rec.709, Display P3, and BT.2020 specifications.

Colour bars

The broadcast test pattern is the gamut triangle enumerated: the three primaries alone (green, red, blue), their pairwise sums (yellow, cyan, magenta), and all three together (white). Every other colour the screen can produce is a mixture of these, inside the triangle.

Five cautions about the xy diagram

- The outline represents standard-observer chromaticity coordinates.
- The page is itself shown through your display’s primaries and calibration.
- Points outside the display gamut can be plotted as locations but not reproduced faithfully.
- Different luminances share the same xy point.
- Equal distances in xy are not equal perceived differences.

Hue, lightness, and chroma do different jobs

Dimension	Useful for	Do not assume
hue	category, identity, highlighting	a natural numeric order
lightness	ordered magnitude if monotonic	equal code steps look equal
chroma	emphasis, status, coarse order	precise quantitative comparison

Let position or length carry values that must be compared precisely. Use colour to organise, order coarsely, or direct attention.

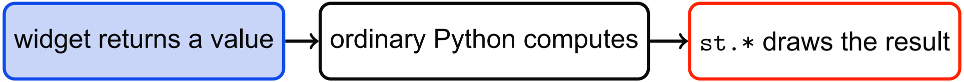
Test a palette in its context

- test colours on the actual background;
- test them at the actual mark and text sizes;
- inspect common colour-vision deficiencies;
- make lightness monotonic in sequential scales;
- reserve diverging palettes for a meaningful midpoint;
- add labels, position, shape, or line style when the distinction matters.

What matters is how the marks look next to each other on the actual page.

Streamlit

How a Streamlit app runs



Whenever a widget changes, Streamlit reruns the script from top to bottom with the new value.

A slider replaces a constant

Notebook or script

```
wavelength = 550
print(wavelength)
```

Streamlit

```
import streamlit as st
wavelength = st.slider(
    "Wavelength (nm)", 380, 700, 550
)
st.write(wavelength)
```

```
python -m streamlit run hello_colour.py
```

The lab builds three linked views

1. **Spectrum**: change peak wavelength and spectral width.
2. **Eye**: integrate the spectrum against three schematic cone curves.
3. **Chromaticity**: use the official CIE table to locate the light in xy.

Final extension: add an intensity slider. Cone-response magnitudes should change; chromaticity should remain almost fixed.

The app's simplifications

- Cone curves in the app are schematic.
- CIE matching functions are standard-observer data, not cone sensitivities.
- A wavelength rendered on screen is an RGB approximation, not monochromatic light.
- xy omits luminance and viewing context.
- The app is an explanatory model, not a calibrated colour-management system.

Exit ticket

Complete the chain for one colour in a chart:

> *The display emits ________________. The eye reduces this to ________________. The viewer perceives it in relation to ________________. The design uses that appearance to encode ________________.*

Name one thing lost at each translation.

Sources and data

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- Streamlit, “Basic concepts” and API documentation.

All plotted figures are course calculations.